

Persistence of Viral Nucleic Acids in Wastewater in Presence of Surfactants

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Motivation

- Wastewater surveillance monitors circulating viral pathogens by quantifying their nucleic acid signals in sewage.
- Surfactants are commonly discharged into wastewater and can degrade nucleic acid signals, potentially biasing surveillance measurements.
- Study objective: Quantify how four surfactant types affect the decay of eight virus targets over 72 hours in influent wastewater.**

Key Takeaways

- For enveloped viruses, signal decay was higher in wastewater than PBS.
- Higher signal decay in the presence of added surfactants occurred almost exclusively in enveloped viruses.
- Ionic surfactants (anionic SDS, cationic DDAC) were more likely to cause higher decay; nonionic Tween 80 had minimal effect.
- Extraviral (free) nucleic acids degraded rapidly in wastewater, with the concentration decreasing by 2 to 4 log₁₀ within 90 minutes.

Results and Findings

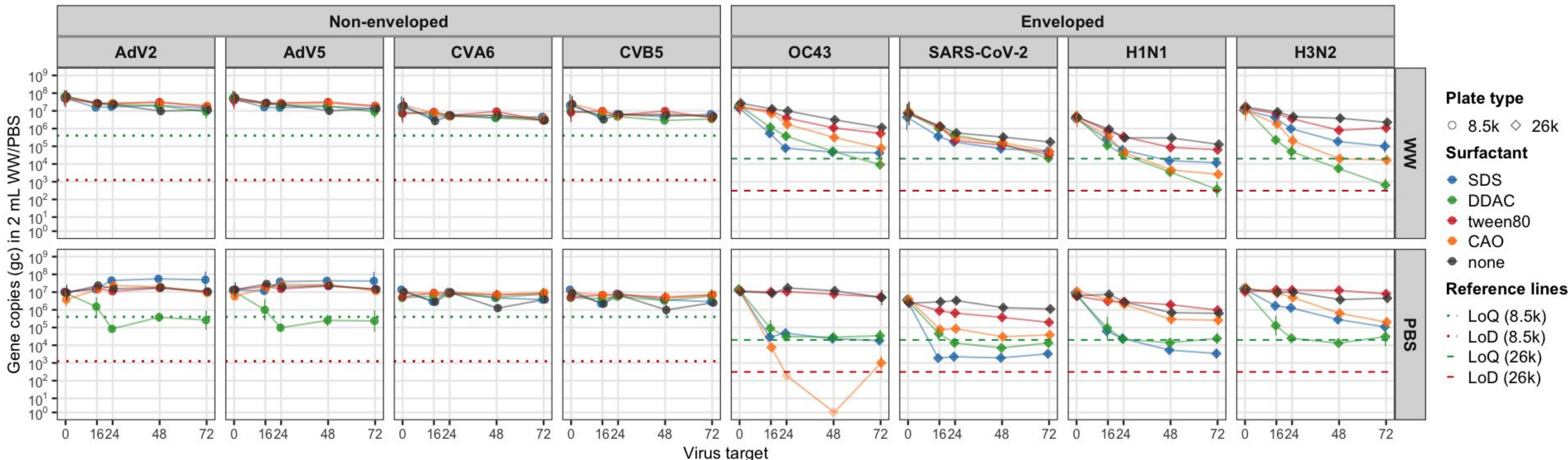


Figure 1: Decay of eight viral nucleic acid targets in wastewater and PBS over 72 hours, with and without added surfactants.

- Nonenveloped viruses showed little to no signal decay in wastewater across all surfactant conditions, except DDAC (AdV2 and AdV5 decay in PBS).
- Enveloped viruses decayed by 2 to 4 log₁₀ over 72 hours in the presence of surfactants.
- For enveloped viruses with no surfactants spiked in, signal decay was more pronounced in wastewater than PBS.

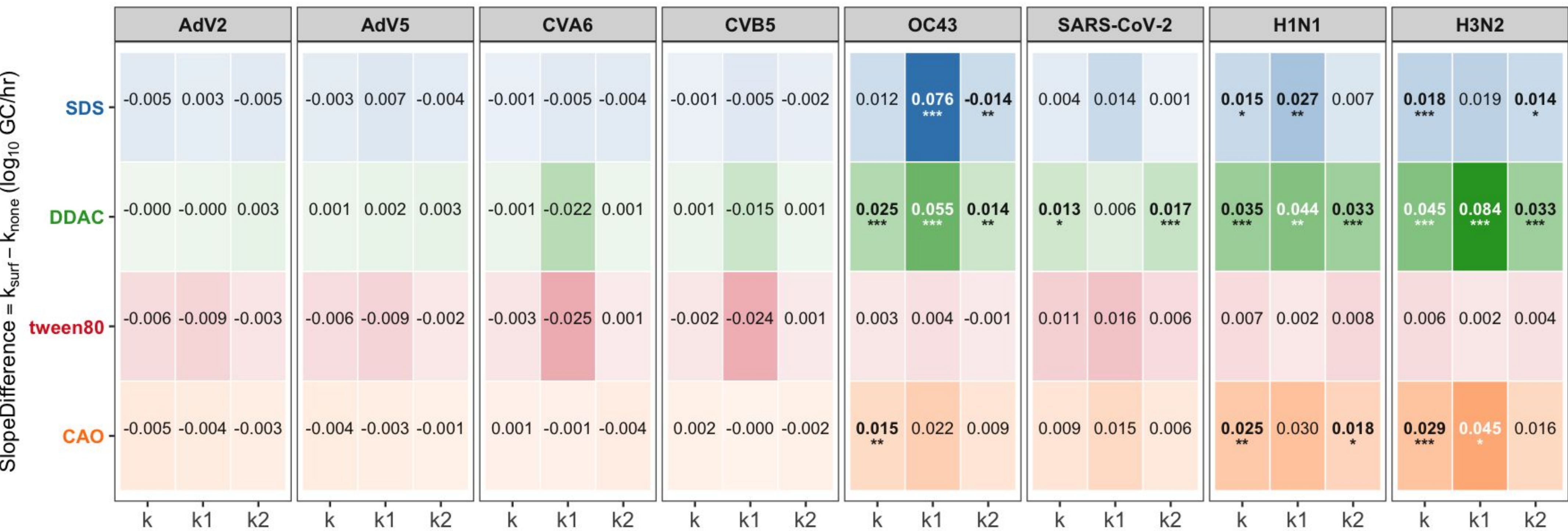


Figure 2: Difference between decay rates with and without surfactant ($k_{surf} - k_{none}$; wastewater only). k = monophasic; k_1 and k_2 = early (0-24 h) and late (24-72 h) biphasic rates.

Stars: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

- Significant surfactant effects were observed for enveloped viruses.
- SDS (anionic) and DDAC (cationic) most consistently caused higher decay rates.
- Analysis using both biphasic (k_1 , k_2) and monophasic decay models provided more holistic understanding.
- Tween 80 (nonionic) had the weakest effects, consistent with surfactant charge driving envelope disruption.

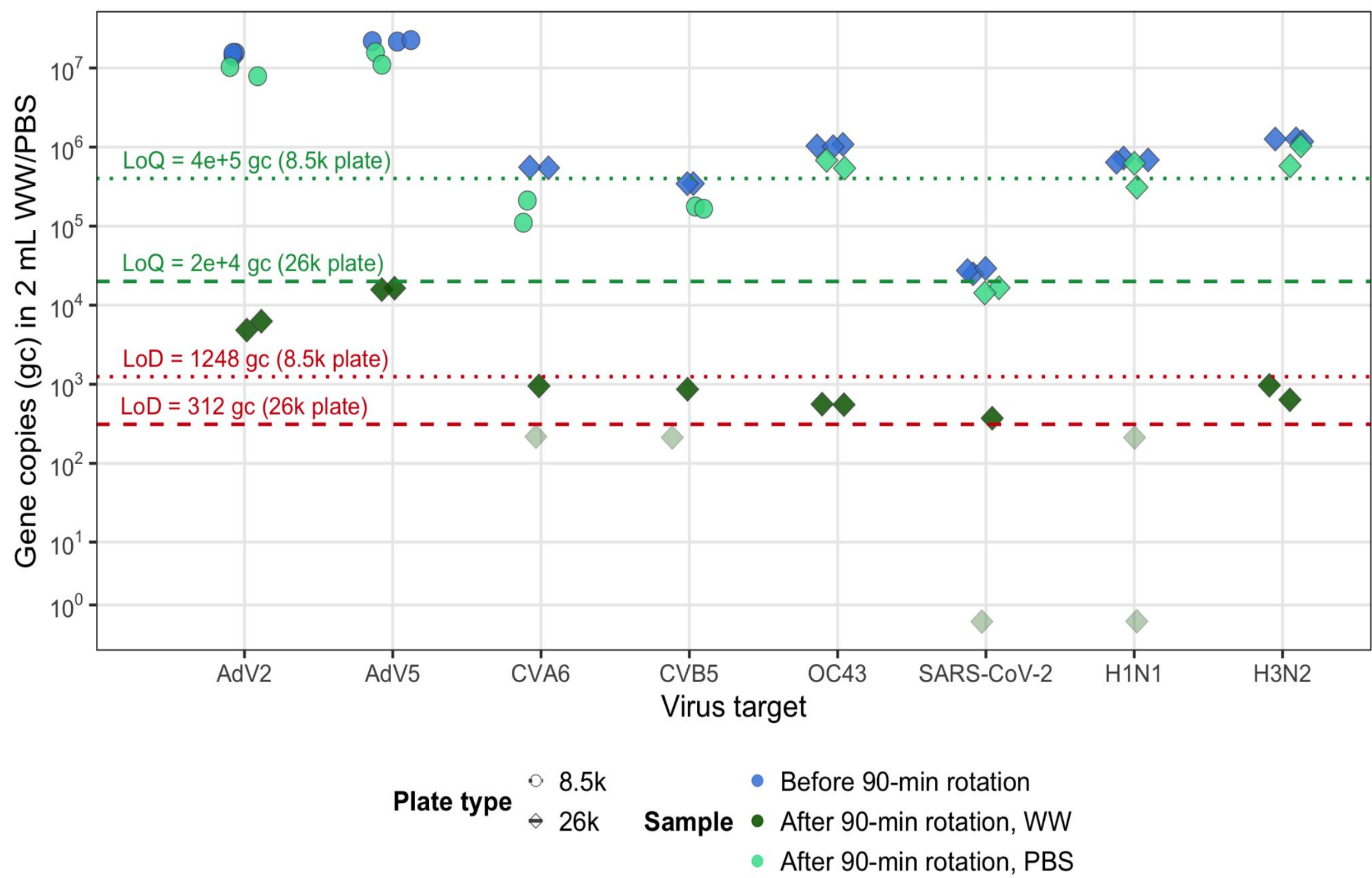


Figure 3: Concentration of extraviral (free) nucleic acid spiked into wastewater before and after 90 min incubation.

- Extraviral nucleic acid degraded rapidly, decreasing 2 to 4 log₁₀ within 90 min.
- This finding confirms that the $t=0$ signal in the decay experiment derived predominantly from intact virus particles rather than free nucleic acids.

Methods

